

SODANKYLA KOKKA, Finland

WMO: 027190

Lat: 67.821N

Lon: 27.746E

Elev: 787

StdP: 14.28

Time Zone: 2.00 (EUE)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.3	(c) -22.1	(d) -33.4	(e) 0.9	(f) -27.3	(g) -27.6	(h) 1.2	(i) -21.9	(j) 25.5	(k) 17.8	(l) 23.1	(m) 18.6	(n) 3.3	(o) 190	(p) 0.771

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	76.6	62.3	72.3	60.0	68.4	58.3	64.4	72.6	62.1	69.7	60.2	66.6	8.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	84.3	67.4	59.1	77.1	64.8	57.2	71.8	62.9	30.0	72.7	28.3	70.2	26.9	66.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.9	(b) 19.7	(c) 17.1	(e) -35.6	(f) 79.3	(g) 3.6	(h) 4.2	(i) -38.2	(j) 82.3	(k) -40.3	(l) 84.8	(m) -42.4	(n) 87.1	(o) -45.0	(p) 90.1		
(4) 22.9	(5) 19.7	(6) 17.1	(7) -35.2	(8) 67.2	(9) 3.8	(10) 3.2	(11) -37.9	(12) 69.6	(13) -40.1	(14) 71.4	(15) -42.3	(16) 73.2	(17) -45.0	(18) 75.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.9	6.9	9.0	15.5	28.6	40.9	50.2	57.1	52.2	44.6	31.0	20.7	13.0
	DBStd	19.85	12.78	15.75	12.06	8.54	8.45	7.31	6.40	6.16	5.79	8.05	11.16	12.78
	HDD50	7438	1337	1147	1069	642	303	85	12	47	180	590	879	1147
	HDD65	12453	1802	1567	1534	1092	748	449	254	398	613	1055	1329	1612
	CDD50	477	0	0	0	0	21	90	232	117	17	0	0	0
	CDD65	16	0	0	0	0	0	4	10	2	0	0	0	0
	CDH74	234	0	0	0	0	5	48	146	35	0	0	0	0
	CDH80	30	0	0	0	0	0	4	19	7	0	0	0	0
Wind	WSAvg	7.8	6.9	7.5	8.5	7.5	8.0	9.4	8.1	7.8	8.0	8.1	7.1	7.2
Precipitation	PrecAvg	22.6	1.5	1.3	1.2	1.3	1.7	2.4	2.9	2.4	2.1	2.0	1.9	1.7
	PrecMax	28.4	2.8	3.0	2.8	3.4	3.3	4.2	5.2	4.9	4.6	4.2	3.1	2.8
	PrecMin	17.6	0.6	0.6	0.3	0.5	0.6	0.7	0.9	0.8	0.3	0.7	0.6	0.4
	PrecStd	3.0	0.6	0.7	0.6	0.6	0.7	0.9	1.1	1.0	1.0	0.9	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	36.4	40.2	52.3	72.8	80.0	82.8	79.4	63.8	48.1	40.6	34.6
		MCWB	31.0	34.8	35.6	43.0	54.8	61.4	63.8	66.1	55.7	46.2	39.1	34.0
	2%	DB	29.1	33.1	36.8	45.5	67.1	72.6	77.8	71.3	60.3	46.1	36.9	32.8
		MCWB	28.7	32.2	33.1	38.4	52.4	59.3	62.8	61.7	53.7	44.2	36.2	32.3
	5%	DB	26.9	31.7	33.9	42.0	60.2	67.5	74.4	67.0	57.4	43.6	34.6	31.7
		MCWB	26.5	31.0	31.5	36.5	48.9	56.4	61.7	58.6	52.1	42.2	34.2	31.3
	10%	DB	24.1	29.2	31.9	39.2	55.0	63.1	70.6	63.6	54.6	41.3	33.0	29.5
		MCWB	23.7	28.6	30.3	34.8	46.8	53.7	59.9	56.9	51.2	40.2	32.6	28.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.2	35.3	35.9	43.3	56.3	64.3	68.4	67.1	57.6	47.8	40.3	34.2
		MCDB	31.5	36.4	39.3	52.3	69.0	75.5	75.0	77.1	61.5	48.1	40.7	34.4
	2%	WB	28.8	32.6	33.8	39.3	53.3	61.4	65.3	62.8	55.1	44.2	36.1	32.5
		MCDB	29.1	33.2	36.2	44.1	64.6	71.1	73.5	69.7	58.8	45.3	36.5	32.8
	5%	WB	26.6	31.0	32.1	37.2	50.6	58.5	63.4	60.1	53.2	42.3	34.2	31.5
		MCDB	26.9	31.5	33.5	41.0	59.7	64.6	71.3	65.0	56.2	43.6	34.4	31.8
	10%	WB	23.7	28.6	30.4	35.6	47.3	55.3	61.5	57.9	51.3	40.3	32.8	28.9
		MCDB	24.1	29.2	31.9	38.3	53.7	61.9	68.5	62.2	54.0	41.3	33.1	29.4
Mean Daily Temperature Range	5% DB	MDBR	16.0	15.6	20.7	16.6	16.4	15.2	17.3	16.5	13.0	9.4	10.9	14.8
		MCDBR	15.9	10.8	14.9	17.8	26.7	23.4	25.8	23.3	19.5	10.0	7.9	9.6
		MCWBR	15.8	10.2	12.6	11.6	14.1	11.3	13.1	13.7	13.2	8.5	7.9	9.6
	5% WB	MCDBR	16.2	10.9	12.0	15.4	25.1	19.2	22.2	20.2	15.6	8.8	7.6	9.6
		MCWBR	16.2	10.4	10.4	10.4	14.2	10.7	12.6	13.1	11.3	8.0	7.6	9.7
Clear-Sky Solar Irradiance	taub		0.193	0.253	0.291	0.321	0.340	0.341	0.354	0.335	0.308	0.272	0.355	N/A
	taud		1.916	2.077	2.147	2.198	2.393	2.471	2.456	2.482	2.536	2.300	2.729	N/A
	Ebn at Noon		36	173	234	261	271	274	266	258	239	170	49	0
	Edh at Noon		3	16	26	33	31	30	29	25	19	12	4	0
All-Sky Solar Radiation	RadAvg		13	163	636	1216	1495	1584	1446	1046	548	179	31	0
	RadStd		1	20	46	77	143	134	154	97	53	32	4	0

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)	+1.10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page